



10

Modeling Disaster Operations Management Problems with System Dynamics

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10.1 Introduction

The aim of this chapter is to illustrate the potential use of System Dynamics (SD) to study Humanitarian operations management (HOM) problems. Before and after a disaster, there are several complex decision-making issues that humanitarian agencies and governments have to deal with in the short, medium, and long term. Managing inflows of population, volunteers, resources, and funds are some examples of complex tasks that humanitarian managers have to address routinely during an emergency. To perform these tasks, managers need to estimate impact at the

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mitigation and preparedness phases, plan service capacity, and make operational decisions during the immediate response phase, among other actions. However, during a disaster, decision-making is subject to uncontrollable and uncertain factors including social behaviors, changing operating conditions, conflicting interests, and unpredictable and interrelated impacts.

Lack of planning and preparation before an emergency, and lack of coordination for immediate response have devastating consequences as observed in the aftermath of hurricane Katrina in 2005 (Farazmand, 2007) or the Haiti earthquake in 2010 (IASC, 2010). Hurricane Katrina evidenced the importance of adequate mitigation and preparedness for disasters. The response when the storm landed was slow and only 80% of the New Orleans population had evacuated when the levees failed (DOT, n.d.). Nearly 90,000 people remained in the city with need of assistance but delivery of supplies was tardy, and there was not a clear plan for receiving, storing and distributing donations (Sebbah, Boukhtouta, Berger, & Berger, 2013).

In the Haiti case, the airport and harbors were not operational immediately after the emergency which created logistical challenges for initial response. Delivery of aid was further delayed for lack of coordination between logistics partners. Humanitarian organizations were overwhelmed by a large influx of volunteers with no previous experience and skills, and of inappropriate in-kind donations (Patrick, 2011). The cholera epidemic that spread after the earthquake evidenced the difficulties of the recovery phase and the importance of reducing vulnerability during recovery and mitigation phases.

System dynamics is a suitable method to analyze complex decisions under dynamic and uncertain conditions, to model the soft variables involved in coordinating multiple decision-makers in uncertain environments, and to anticipate the unintended consequences of actions. The system thinking approach of SD helps with problem formulation by identifying cause and effect relationships and feedback cycles. SD models built from these relationships are useful for understanding accumulation processes and complex dynamic behaviors and once validated, for generating scenarios and testing alternative policies.

As discussed in following sections SD has been applied to study a variety of HOM problems such as: service capacity and delivery of humanitarian supplies, customs clearance procedures, resource allocation (e.g. volunteer and personnel training, hiring personnel, trucks, fleet vehicles), and staff learning and training process.

This chapter is organized as follows: The next section describes the humanitarian operations (HO) associated with the phases of the disaster management cycle. Section 10.3 discusses the complexity of decision-making during disasters, while Sect. 10.4 presents a summary of OR/MS methods for supporting decision-making in HOM. Section 10.5 describes the system dynamics methodology and Sect. 10.6 illustrates the application of an SD model for understanding a simple HO problem.

10.2 Humanitarian Operations Management and the Disaster Cycle

Natural and man-made disasters endanger or devastate human life, properties and environment (Nikbakhsh & Zanjirani Farahani, 2011). Humanitarian relief operations seek to reduce loss of life, ease suffering, and advance human dignity through research on planning and responding to disasters (Ergun, Keskinocak, & Swann, 2011; Van Wassenhove, 2006). Four phases: mitigation, preparedness, response and recovery make up the disaster management cycle (Altay & Green, 2006; Van Wassenhove, 2006).

10.2.1 Mitigation

Mitigation and preparedness take place before disaster events. Mitigation seeks to prevent hazards or reduce the impact of disasters through long-term planning and investments. Structural mitigation measures apply technological advances while nonstructural measures like legislation, land-use planning and insurance create conditions for preventing disasters and reducing risk (Nikbakhsh & Zanjirani Farahani, 2011).

10.2.2 Preparedness

Preparedness involves making plans to respond to a disaster and ensuring that five key elements: human and financial resources, knowledge and process management, and community are in place (Van Wassenhove, 2006). These elements are interrelated as explained by Van Wassenhove (2006).

Selection and training of human resources (HR) helps to develop planning, coordination and execution capabilities in local and international teams. Knowledge management, which consists of capturing, codifying and transferring operations knowledge about operations learned from previous disasters, aids in developing HR capabilities. Operations and process management involve logistics decisions such as setting up goods, agreements and means to move resources quickly. Operations can be improved by developing agreements with other humanitarian organizations, governments, military, and business (community). Finally, sufficient funds and financial resources allow preparation and a smooth response.

10.2.3 Response

In the response phase resources, personnel and equipment are immediately dispatched to the disaster area and emergency procedures are employed. In this phase, the quality and speed of logistics impacts the ability of local communities to reconstruct themselves after a disaster (Leiras et al., 2014) and successful response requires a plan for coordination of relief forces and operations (Altay & Green, 2006).

10.2.4 Recovery

Finally, recovery includes rehabilitation and reconstruction activities to restore affected areas to their previous state (Van Wassenhove, 2006), as well as community self-sufficiency and local/national governance (Goldschmidt & Kumar, 2016). During recovery there are opportunities to reduce future risks by implementing mitigation techniques while, ideally, lessons learned after the disaster inform mitigation and preparedness phases (Ergun et al., 2011; Kovács & Spens, 2009; Zeimpekis, Minis, & Ichoua, 2013).

10.3 Decision Context in Humanitarian Operations Management

HOM deals with unpredictable impacts of disasters, complex operating conditions and changing needs (Overstreet, Hall, Hanna, & Kelly Rainer, 2011; Van Wassenhove, 2006). HO decisions depend on environmental, economic and social contexts (Kunz & Reiner, 2012) and decision-making in each part of the disaster cycle involves multiple parties and organizations with sometimes conflicting objectives. Interaction of local and international humanitarian organizations (IHO), private sector companies, governments, military, and individuals is complex, as these organizations face uncertain and dynamic conditions, time pressure, resource constraints, and rigid funding structures (Besiou, Stapleton, & Van Wassenhove, 2011; Kovács & Spens, 2009; Van Wassenhove, 2006).

Van Wassenhove (2006) highlights how diverse and interacting factors hide the nature of problems, and increase complexity of disaster operation management. Ignored factors, and ambiguous cause-effect relationships further complicate disaster management. Multiple feedback loops arise as bad crisis management triggers new phenomena and actors and operations interact.

10.4 Research for Decision Support in HOM

HOM is an emerging research field in operations management (OM) and supply chain management (SCM) that has gained increasing attention of many OM researchers (Altay & Green, 2006; Ergun et al., 2011; Goldschmidt & Kumar, 2016; Kovács & Spens, 2009).

Most studies in HOM focus on the response, preparedness and mitigation phases of a disaster (Altay & Green, 2006; Kunz & Reiner, 2012). Despite its importance in mitigating risks (Goldschmidt & Kumar, 2016), the recovery phase is less studied (Kunz & Reiner, 2012).

OR/MS methods, including decision analysis, mathematical programming, expert systems and simulation, have been applied to support decision-making in HO. (Kunz & Reiner, 2012; Simpson & Hancock,

2009; Overstreet et al., 2011; Özdamar & Ertem, 2015; Kovács & Spens, 2011). Planning of delivering systems and inventory control during the response and preparedness phase are the most commonly addressed problems (Kunz & Reiner, 2012; Goldschmidt & Kumar, 2016). Facility location, inventory management, and network flows problems are formulated and solved using mathematical programming and heuristics while uncertain disaster impacts are studied using stochastic programming and scenario analysis models (Altay & Green, 2006; Leiras et al., 2014; Overstreet et al., 2011; Özdamar & Ertem, 2015).

Mathematical programming models and the normative solutions they provide rely on data and strong assumptions about resources availability. Thus, simulation methods, including discrete event, agent-based and SD models, have been applied to manage uncertainty and incomplete knowledge characterizing HOM (Altay & Green, 2006).

Among the applications of simulation to HOM are traffic (Özdamar & Ertem, 2015) and mass evacuation (Swamy, Kang, Battam, & Chung, 2016), understanding of chain events (Santella, Steinberg, & Parks, 2009), and training for understanding the complexity of resources planning (Albores & Shaw, 2008; Rauner, Schaffhauser-Linzatti, & Niessner, 2012). SD and its applications for strategic planning and complex systems analysis are underrepresented in HO literature (Altay & Green, 2006).

Table 10.1 summarizes the range of applications of SD in HOM. These applications include systems thinking for problem formulation, generic models for learning and understanding complex dynamics issues and models for supporting strategic decision-making. Most SD applications in Table 10.1 relate to planning for the preparedness and mitigation phases of the disaster cycle.

Planning for human-made and natural disasters is, in some aspects, a wicked problem as each disaster is unique and its impacts and solutions depend on social conditions and political decisions. IHO's, for instance, need to coordinate their actions with multiple stakeholders from private sector companies to governments and donors in order to assist victims of human-made and natural disasters that develop in complex environments (Besiou et al., 2011). Systems thinking and SD has been useful for identifying causal relationships and structuring complex problems such as coordination of partnerships (Heaslip, 2013) or the vicious cycle

Table 10.1 Applications of system dynamics and systems thinking to HOM

Systems thinking	
<i>Problem structuring and understanding</i>	
Mitigation	Causal diagrams are used to represent causal relationships between system components and to identify the disruptive variables and feedback mechanisms that can amplify disruptions, thus creating risks (Powell, Mustafee, Chen, & Hammond, 2016). A system dynamics framework is used to propose and perform econometrics tests of causal and feedback relationships between disasters occurrence and economic and population vulnerability (Sodhi, 2016).
Preparedness	Dynamics of risk potential in mass-gathering events and its implications for policy (Handel, Biedermann, Kielar, & Borrmann, 2014).
Response	Coordination and partnership selection in logistics operations involving military and aid organizations. Uses systems analysis and design techniques (SADT) and system dynamics for identifying causal relationships (Heaslip, 2013) Flow of financial resources from different donors and with different constraints (Burkart, Besiou, & Wakolbinger, 2016)
Generic models for learning and understanding dynamic complexity	
<i>Feedback loops and delays</i>	
<i>Uncertainty</i>	
<i>Accumulation dynamics</i>	
Mitigation	Models the relationship between the quantity of non-novel events and organizational crises. The rate of interruptions resolution decreases as incoming interruptions arrive and accumulate (Rudolph & Repenning, 2002).
Preparedness	A model for the 1992 Westrey mine disaster in Nova Scotia, Canada shows that accidents are outputs of a complex system in which organizational and personal commitment is needed to create a safety culture (Cooke, 2003). State of preparedness increases with technology changes, attitudes to prevent risk, and resource state availability (Kwesi-Buor, Menachof, & Talas, 2016).
Response	Model for the evacuation process includes evacuees and relief units conducting evacuation and installing mobile protections during a flood (Berariu, Fikar, Gronalt, & Hirsch, 2016).
Recovery	Labor availability limits the ability for housing reconstruction after a disaster. Labor management model to support housing construction after a disaster (Kumar, Diaz, Behr, & Toba, 2015). Flow of materials during reconstruction (Diaz, Kumar, & Behr, 2015).

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Table 10.1 (continued)

	Models for supporting strategic decisions
	<i>Resource allocation</i>
	<i>Capacity building</i>
	<i>Impact of capacity and information delays on lead times</i>
Mitigation	Flood relief costs in the U.S. continue to rise, despite the availability of federal policies. A generic system dynamics model is developed to show how policies can increase vulnerability in flood-prone areas (Deegan, 2006). The Critical Infrastructure Protection Decision Support System (CIPDSS) is a system dynamics model of the interdependences between critical infrastructures in a disaster. A scenario for road and telecommunications damage is examined (Santella et al., 2009).
Preparedness	Mitigation and preparation for urban disasters, focusing on preparation of urban subsystems: population, industry, transportation, public facilities, housing, life-support and environmental protection (Ho, Chienhao, & Hsiao-Lin, 2006). A model of field vehicle fleet management for international humanitarian organizations is developed and applied to study the impact of donations and media attention on humanitarian relief operations (Besiou et al., 2011) Field vehicle management in international humanitarian organization. SD model for long-term planning of 4WD vehicles under management and funding scenarios (Besiou, Pedraza-Martinez, & Van Wassenhove, 2014)
Response	The delivery of ready to use therapeutic food items for immediate response phase is modeled using system dynamics. Scenarios of developing disaster management capabilities and prepositioning inventory are compared. The impact of developing management capabilities in custom clearance lead times is estimated (Kunz, Reiner, & Gold, 2014). Impact of dynamic road conditions and information delays on on-time transportation rate (Min Peng, Peng, & Chen, 2014). Dynamic lead-time prediction after an earthquake considering dynamic road conditions and information delays (M Peng, Chen, & Zhou, 2014).
Recovery	Identify temporary debris management sites and evaluate debris removal under different scenarios using SD and GIS (Kim, Deshmukh, & Hastak, 2018).

(continued)

Table 10.1 (continued)

Models for policy support	
	<i>Policy evaluation</i>
	<i>Potential impacts evaluation</i>
Preparedness	Social and psychological factors, including awareness of risk and knowledge of disasters, affect population acceptances of evacuation orders (Simonovic & Ahmad, 2005).
	Disaster planning in remote rural communities, specifically hospital's surge capacity (Hoard et al., 2005).

linking disaster impacts with increased economic and population vulnerability (Sodhi, 2016).

Regarding dynamic complexity, generic SD models have been developed to increase learning and understanding of the effect of feedback loops, delays, and accumulation dynamics in HO such as evacuees and resources flows (Berariu et al., 2016; Diaz et al., 2015; Kumar et al., 2015) and on risk (Cooke, 2003; Rudolph & Repenning, 2002) and preparedness (Kwesi-Buor et al., 2016).

Other SD models applications aim at supporting strategic decisions, such as capabilities development, and resources allocation, considering information and transportation delays. Finally, other models are developed to analyze operative policies affecting evacuations (Simonovic & Ahmad, 2005) and hospital surge capacity (Hoard et al., 2005).

10.5 System Dynamics in HO

SD is a policy-oriented quantitative systems approach classified within the problem-structuring methods and soft systems approaches. It emerged as a system modeling and simulation methodology to capture, understand, and analyze the dynamics of complex systems. The system complexity is determined by non-linear interactions among the elements of the system, multiple feedback processes, time delays, and stock and flow structures (Sterman, 2000).

Feedback means that problems are consequences of our previous decisions and actions. That is, assuming that managers see problems as discrepancies between the current situation and their goals, solving these problems entails decisions leading to actions and results that change the current situation alleviating the current problem, but generating new discrepancies with our next goals (Morecroft, 2015). Feedback processes are represented by closed loops formed by the interaction among variables and describe the way a decision can determine the state of the system in the short or long term. In feedback processes, a change in a variable leads either to counteracting (i.e., balancing or negative feedback loop) or reinforcing (i.e., reinforcing or positive feedback loop) the change when the effects of that change are followed around the loop (Morecroft, 2015). The dynamics of the system arises from the interaction between feedback loops.

Feedback is often delayed. Time delays between actions and their impact on the state of the system make long-term responses different from short-term responses and increase dynamic complexity (Sterman, 2000). As discussed by Sterman (2000) time delays reduce our ability to accumulate experience by cycling around learning loops, and cause interventions to continue well after they are needed, producing instability and oscillations.

Managers usually make decisions based on their experience and intuition considering problems as event-oriented systems, even in complex situations, and ignoring feedback (Sterman, 2000). The SD methodology facilitates the comprehension of problems and allows gaining relevant insights into the feedback processes which determine the system behavior.

A logical and intuitive process is used to capture key complexities of a system (Kumar et al., 2015) and analysis tools allow managers to evaluate the short and long-term effects of their decisions. SD is useful to learn about system complexity, understand the source of policy resistance, and design better operating policies to improve systems' performance (Sterman, 2000). SD is also used as a systems thinking tool for analyzing HOM problems as complex systems formed by interconnected elements with the same purpose (e.g., Powell et al., 2016). Through SD we can construct broad-brush models to capture a holistic view of the system and understand cause-and-effect relationships.

10.5.1 System Dynamics Modeling Process

System dynamics modeling is a continuous and iterative learning process which involves articulating the problem, formulating a dynamic hypothesis, building, testing, and reviewing a simulation model, and designing and evaluating policies (Sterman, 2000). Problem articulation implies defining the issue under investigation, the key variables and factors that must be considered to address such issues, the time horizon, and the reference modes (i.e., the historical and future behavior of key variables). Stakeholders associated with the problem play an important role in this step. During problem articulation, the modeling team must keep in touch with stakeholders to gather information about the issues that concern them the most and their purpose with the model (Sterman, 2000). The modeling team must capture enough information for building a model aimed at addressing a specific problem rather than an entire system.

10.5.1.1 Dynamic Hypothesis Formulation

The dynamic hypothesis is a preliminary theory, developed by the modeling team together with the client, to account for the structure that yields the system behavior through time. This hypothesis provides the modeling team with an endogenous explanation for the behavioral patterns of the system generated by the interaction of the variables (Sterman, 2000). SD offers a set of diagramming tools to facilitate developing and understanding this hypothesis. The most used are causal loop diagrams (CLDs) and stock and flow (S&F) maps.

10.5.1.2 Causal Loop Diagrams

A CLD is a flexible and visual tool useful for capturing our mental models about the dynamic hypothesis of the system by diagramming the cause and effect relationships between variables that depict the feedback structure (Morecroft, 2015; Sterman, 2000). A CLD is drawn by connecting variables through links (arrows), indicating that a variable has influence

on another. Each link is assigned a polarity, either positive or negative, to denote the effect the causing variable has on the influenced variable. A link with positive ‘+’ polarity implies that effect moves in the same direction than the cause (i.e., as the cause increases, the effect increases as well; or, as the cause decreases, the effect decreases too), whereas a negative ‘-’ polarity entails that effect moves in the opposite direction than the cause (i.e., as the cause increases, the effect decreases; or, as the cause decreases, the effect increases). Feedback loops are easily identified in a CLD by tracing the effect of a change in a variable. We have a feedback loop when that effect propagates across the arrows linking the variables and arrives at the variable which originally effects the change. If that effect strengthens the original change, there is a positive (i.e., reinforcing) feedback loop; otherwise, if the effect contrasts with the original change, there is a negative (i.e., balancing) feedback loop (Sterman, 2000).

10.5.1.3 Stock and Flows Maps

CLDs are very useful for capturing mental models and communicating feedback structure. However, they do not differentiate between the S&F structure of dynamic systems. Complex problems are modeled using SD under a S&F structure where stocks represent state variables that describe the state of the system and change over time depending on the flows. Subjects’ decisions are based on information generated by the stocks and the resulting actions are reflected in the flow rates.

In an S&F map, tanks (rectangles) represent stocks, while pipes controlled by valves represent flows. Flows can be inflow or outflow variables. Inflow and outflow variables control stock levels by increasing and decreasing them. Tangible (e.g., material and goods) and intangible (e.g., information) items flow through S&F structures. Stocks perform as a kind of memory which stores the results of past decisions (Morecroft, 2015). For example, stocks can represent inventory levels, people waiting to be evacuated, a backlog of patients, humanitarian supplies waiting for the clearance certificate and warehouse capacity levels, among others. Inflow variables include incoming humanitarian aid, allocation rate of

relief units, injured people and capacity-acquisition rate, among others. Outflow variables could be humanitarian aid delivery rate, surgery rate, medical service rate and evacuation rate, among others. Flows originating outside the boundary of the model surge from a source, while flows leaving the model drain into a sink. Both sources and sinks are represented by a cloud (Sterman, 2000).

In addition to the stock and flow variables, a SD model contains auxiliary variables. Such variables enable different S&F structures communicate among them. These can be endogenous (i.e., as functions of stocks or other auxiliary variables) or exogenous variables (i.e., parameters or constants).

10.5.1.4 Formalization of the Model

The model requires equation formulation, parameterization and simulation for understanding the structure/behavior relationship (Lane & Husemann, 2008). Equation (10.1) is the mathematical expression for a S&F structure. This equation determines the way stock variables accumulate according to the net flow—inflow rate minus outflow rate.

$$\begin{aligned} \text{Stock level}(t) = & \int_{t_0}^t [\text{Inflow}(t) - \text{Outflow}(t)] dt \\ & + \text{Stock level}(t_0) \end{aligned} \quad (10.1)$$

The S&F's structure and equations allow us to transform the qualitative feedback loops of the CLD into quantitative dynamic models useful for diagnosing performance problems, testing policies and strategies, and designing experiments for learning from the complexity of the systems (Morecroft, 2015).

The next step after clearly defining the structure of the system behavior is to formalize it by building a computer simulation model. Building a simulation model includes specifying its structure in a computer program,

allocating constant values to the exogenous variables (parametrization) and validating or testing the model for consistency within its purpose and boundaries (Sterman, 2000). The computer simulation models are used to conduct experiments to test the dynamic hypothesis and assess policies and strategies. To test the dynamic hypothesis, simulated behavior is compared to the actual behavior of the system.

Once built, the formal computer simulation model must be validated by applying different tests described by Sterman (2000) and Barlas (1994). The objective of validation is to establish that the model satisfies its intended purpose. Validation tests include running the model under extreme conditions, verifying that it obeys physical laws, checking dimensional consistency, assessing parameters, performing sensitivity analysis and testing integration errors, among others. For a detailed description of validation tests see Sterman (2000).

10.5.1.5 Policy Analysis

SD methodology allows decision-makers to create alternative what-if scenarios to assess their action policies without taking risks in the real system. When the simulation model is finished and validated, the modeling team interacts with stakeholders to establish a common framework for policy analysis (Hoard et al., 2005). The policy analysis provides decision-makers with a picture of how their ideas or strategies might perform, and enables them to design the best path to success. In the SD context, policies refer to strategies, new structures and decision rules aimed at improving the system behavior. That is, policies can involve changing or adding feedback loops, eliminating time delays, modifying the decisions-making process, and/or changing parameters (Sterman, 2000). By carrying out a policy analysis with a SD model, decision-makers can obtain evidence of how their strategies impact the system performance without losing sight of the control variables of the system and avoiding risks.

10.6 An Illustrative Example: Water Supply after a Disaster

The supply of water to the population affected by a disaster is a relevant dynamic issue during the response phase. Figure 10.1 illustrates a simple S&F diagram for describing the process of delivering and distributing water to an affected community. Tankers deliver water from a nearby location and volunteers in the disaster zone distribute water to the affected population. Two key resources are modeled: tankers and volunteers. A number of tankers and volunteers are available in the first response stage, and the model finds the additional tankers and volunteers needed to deliver water within desired standards.

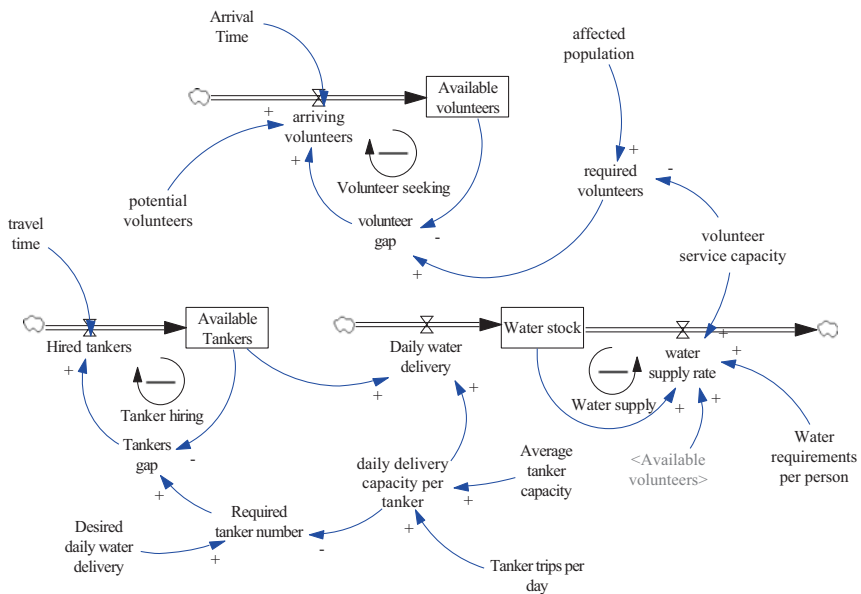


Fig. 10.1 Stock and flow diagram for water delivery after a disaster

10.6.1 The Dynamic Hypothesis of the Problem

Each tanker has a *daily delivery capacity* that depends on its capacity and the number of trips it makes during the day. All tankers are assumed to have the same *average tanker capacity* and make the same number of trips per day (*tanker trips per day*). The variable *required tankers number* is a function of the *desired daily water delivery*, which represents the estimated cubic meters (m^3) per day needed to attend to the community and is assumed to be constant, and of the *daily delivery capacity per tanker*. The *required tankers number* is the goal of the balancing loop *tanker hiring*: Tankers are hired to fill the gap between available and required tankers, and their arrival is delayed by a *travel time*.

The water the tankers deliver is collected for distribution to the population, depending on the service capacity of the available volunteers. *Daily water delivery* depends on the number of *available tankers* and their *daily delivery capacity*. Volunteers distribute water at a *water supply rate* that depends on the number of *available volunteers*, *volunteer service capacity* and *water requirements per person*.

A balancing loop called “*volunteer seeking*” represents the process of finding volunteers to reach a distribution goal. The number of required volunteers depends on the affected population and on volunteer service capacity. A volunteer gap is defined as the difference between available and required volunteers, and potential volunteers are called and arrive to fill this gap.

10.6.2 Formalization of the Model

Equations (10.2), (10.3), and (10.4) are the mathematical expressions of the S&F structure and determine the dynamics of the system (Sternan, 2000). They represent the way *available tankers*, *available volunteers*, and *supplied water*, respectively, accumulate according to their net flow—inflow rate minus outflow rate. An equivalent expression is Eq. (10.5), which describes the net change in the water stock at any time t .

$$\begin{aligned} \text{Available tankers}(t) = & \int_{t_0}^t [\text{Hired tankers}(t)] dt \\ & + \text{Available tankers}(t_0) \end{aligned} \quad (10.2)$$

$$\begin{aligned} \text{Available volunteers}(t) = & \int_{t_0}^t [\text{Arriving volunteers}(t)] dt \\ & + \text{Available volunteers}(t_0) \end{aligned} \quad (10.3)$$

$$\begin{aligned} \text{Water stock}(t) = & \int_{t_0}^t [\text{Daily water delivery}(t) \\ & - \text{Water supply rate}(t)] dt \\ & + \text{water stock}(t_0) \end{aligned} \quad (10.4)$$

$$\begin{aligned} \frac{d\text{Supplied water}}{dt} = & \text{Daily water supply}(t) \\ & - \text{Serving rate}(t) \end{aligned} \quad (10.5)$$

Model parameters for this example are based on water delivery recommendations by the World Health Organization (WHO & WEDC, 2011, 2014) and own assumptions, and summarized in Table 10.2.

10.6.3 Simulation Results

The dynamic issue at the water supply problem is coordinating supply capacity given by available tankers and available volunteers with the water requirements of the community. Figure 10.1 shows that the dynamics of both the available tankers and available volunteers is determined by negative feedback loops which act to bring the states of these two stock variables in line with the desired states. Figures 10.2a and b illustrate the resulting behaviors of this negative feedback structure on the available tankers and the available volunteers. Figure 10.2c shows that the water stock increases. This occurs because the *desired daily water delivery* (200 m³/day) is higher than the *desired water supply rate* (150 m³/day)

Table 10.2 Parameters for model formalization

Estimated parameter	Value	Units	Sources/assumptions
Desired daily water delivery	200	m ³ per day	Based on (WHO & WEDC, 2014)
Average tanker capacity	5	m ³	Based on (WHO & WEDC, 2014)
Tanker trips per day	11	Trips per day	Based on 130 minutes gross turnaround time in (WHO & WEDC, 2014)
Water requirements per person	0.015	m ³ per day	Based on 15.2 liters per day requirements for a community during survival phase in (WHO & WEDC, 2011)
Arrival time (volunteers)	2	Days	Assumed
Travel time (trucks)	2	Days	Assumed
Affected population	10,000	Persons	Assumed
Volunteer service capacity	96	Persons/ (volunteer-day)	Based on assumption of 5 minutes for supplying water to affected persons

and the gap between the desired and actual available tankers is corrected faster than the gap between the desired and actual available volunteers, as shown in Figs. 10.2a and b. Enough tankers and budget are available to satisfy hiring needs while the arrival of new volunteers depends on whether potential volunteers are willing to travel to the disaster site. Consequently, the success of water delivery depends on how quickly potential volunteers can be engaged. According to Fig. 10.2d, it takes about 17 days for the system to reach the desired water supply. Thus, our analysis suggests that design policies should be focused on attracting new volunteers.

10.6.4 Policy Analysis

The model is used to test an advertising policy to attract potential volunteers. We assume that by advertising volunteer campaigns we can increase,

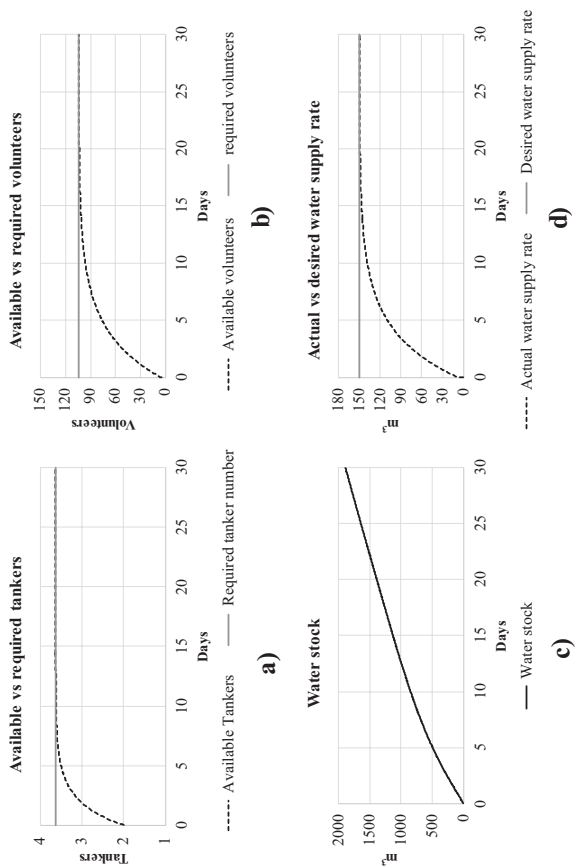


Fig. 10.2 Base case simulation results

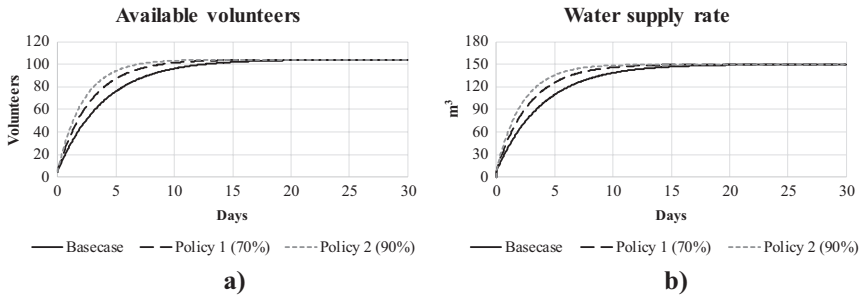


Fig. 10.3 Policy analysis results

to some extent, the impact of potential volunteers on the net inflow volunteers (i.e., arriving volunteers). To illustrate the impact of such a policy, we change the parameter of potential volunteers and again simulate the model with the new values. This parameter represents the effectiveness of campaigns to attract and train new volunteers and is modeled as a percentage of the volunteer gap. Figure 10.3 illustrates the resulting behavior for three different values of the parameter *potential volunteers*: 0.5 (base case), 0.7 (policy 1), and 0.9 (policy 1). Results show that the higher the share of potential volunteers reached by campaigns, the faster the new volunteers arrive, and the quicker the target water supply is achieved. This result reflects the importance of the number of available volunteers for appropriately attending to a catastrophe.

10.7 Conclusions and Future Work

This chapter illustrates how SD can be used as a framework to study HOM problems. The SD simulation and modeling process has been described, as have the kind of HOM issues that can be addressed. Learning is one of the main applications of simulation and SD models in HO. Learning through models allows for continuous improvement of disaster operations management from governments and humanitarian organizations.

SD modeling and simulation provides humanitarian decision-makers with information to make strategic decisions that increase the performance

of the entire humanitarian supply chain. By modeling feedback mechanisms, accumulations and delays, system dynamics helps governments to understand the unintended impacts of their policies and to be aware of the long-term performance of plans for mitigating disasters and protecting populations from hazards. Disaster planning includes strategies for vulnerability reduction. These strategies can be tested by simulating SD models under different impact and exposure scenarios that capture disasters' uncertainty and without risking actual resources and lives. SD simulation can also help to reduce risks by educating people about the hazards and threats posed by natural disasters.

SD modeling has the potential to contribute distinctively to HOM issues. It is a useful methodology to simulate a plethora of scenarios to learn from the results and develop strategies and alternative structures for the improvement of the HOM. The humanitarian research community has mainly focused on studying issues at the response and preparedness phases, while the recovery phase is largely unexplored. Actions in the recovery phase are key to reduce risk exposure and vulnerability, but their impact depends on complex social and economic conditions. For its ability to model soft variables of decision environments, SD is a useful tool to devise reconstruction strategies and assess their long-term impacts, especially if combined with other simulation techniques and GIS. SD will be a powerful tool to explore the long-term impacts caused by strategies aimed at developing local capabilities for responding to emergencies.

Integration of SD and AB, DES, and SIG modeling and simulation is a promising area of research. Integration of GIS and SD can, among other applications, model the relationships between the physical, economic and social vulnerability of communities, and explain the interactions between natural systems (e.g., wildland, coastal) and urban areas. Neuwirth, Peck, and Simonovic (2015) propose a framework for using GIS data as initial values for the stock variables of a spatially disaggregated SD model. Modeling of spatial and temporal feedback is proposed for assessing flood risks (Ahmad & Simonovic, 2015), and coastal storm damages (Hartt, 2014). The applications of SD and SIG integrated modeling are not limited to the mitigation phase. For the recovery phase, the integration of spatial information and system dynamics has

been proposed to find locations and design resilient debris management systems (Kim et al., 2018).

Some HOM problems such as the mass movement or convergence of people, supplies and information have the structure of a queuing system in which servers (volunteers), donations and messages arrive and are assigned to serve arriving customers (victims) (Lodree & Davis, 2016). Important assumptions of queuing models, such as considering arrival and service rates as exogenously determined and the steady-state condition (i.e., arrival rates do not exceed, on average, service rates), are not usually satisfied in disaster conditions. SD allows for relaxing queuing models' steady-state assumptions, and for modeling arrival and service rates as the result of endogenous processes and as influenced by soft variables such as media coverage and individual impulses.

Finally, many organizations with conflicting objectives interact in humanitarian operations. A combination of system dynamics with multi-criteria decision analysis is proposed to capture decision-makers' priorities and to design policies to achieve several objectives (Pruyt, 2007; Tsaples, Papathanasiou, & Ploskas, 2017)

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